

Computational Chemistry

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Textbook:

1. Christopher J. Cramer, Essentials of Computational Chemistry: Theories and Models, John Wiley & Sons, LTD, New York, 2002
2. F. Jensen, Introduction to Computational Chemistry, John Wiley and Sons, New York, 2001

What is Computational Chemistry?

- ❖ Computational Chemistry studies the structures of atoms, molecules and crystals, and explores the relationship between properties and structures.
- ❖ Computational chemistry is one of the most rapidly advancing and exciting fields in the natural sciences today.
- ❖ Computational chemistry has become widely used because of Dramatic increase in computer speed and the Design of efficient quantum chemical algorithms.

Robert S. Mulliken

1966 Nobel Prize in Chemistry



"...the era of computing chemists, when hundreds if not thousands of chemists will go to the computing machine instead of the laboratory...is already at hand. There is only one obstacle, namely, that someone must pay for the computing time."

Overview of Computational Chemistry

Major goals

- To develop efficient mathematical approximations and computer programs
- To apply these programs for calculating the electronic structure and properties of chemical systems.
- To get insight into chemical phenomena.

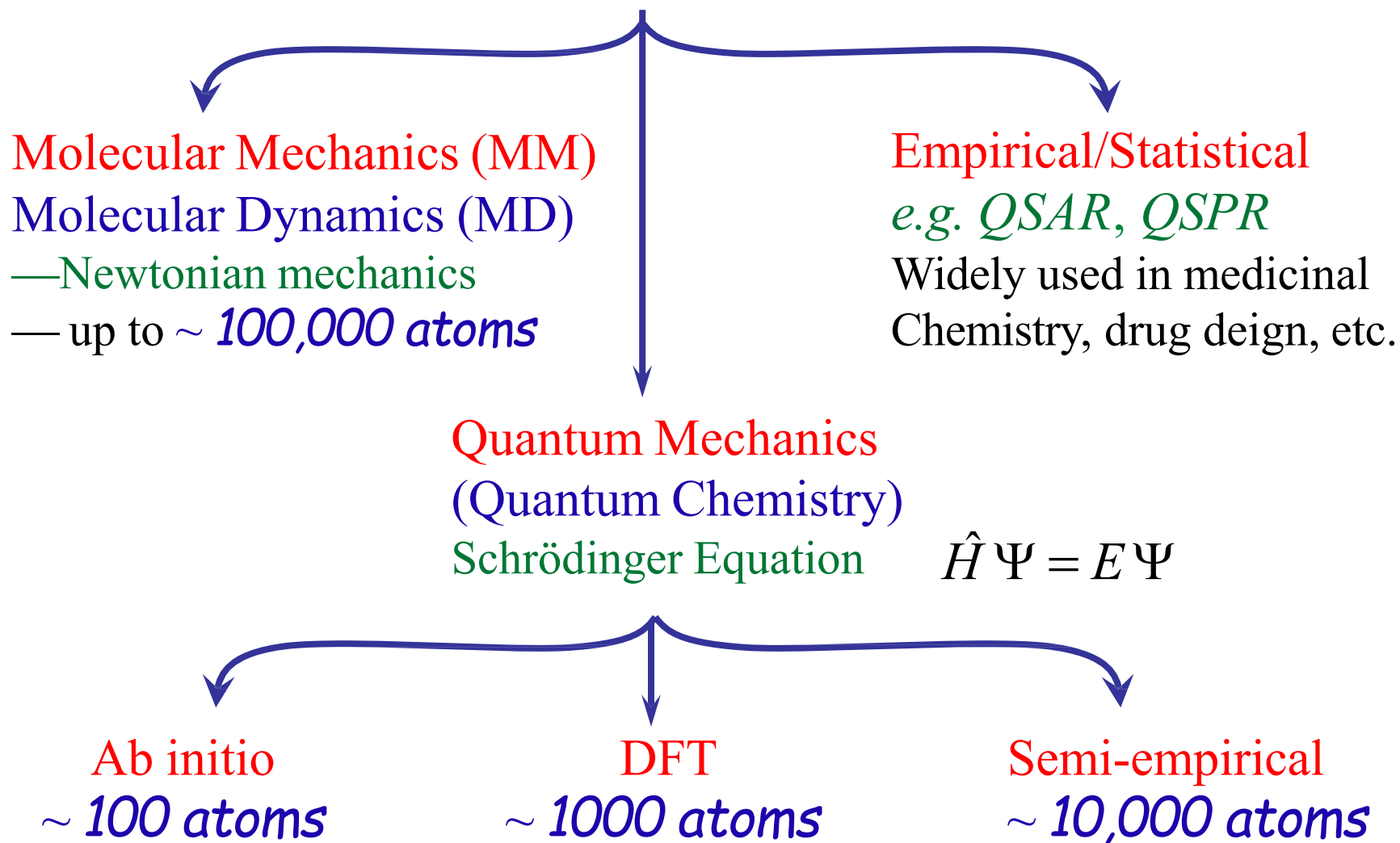
Electronic structure calculations are useful for:

- Predicting structure and properties (new materials),
- Calculate structures for highly reactive species,
- Confirming and explaining experimental data,
- Understanding and predicting chemical phenomena,
- Modelling the biochemical compounds,
- Modelling the intermolecular interactions, etc.

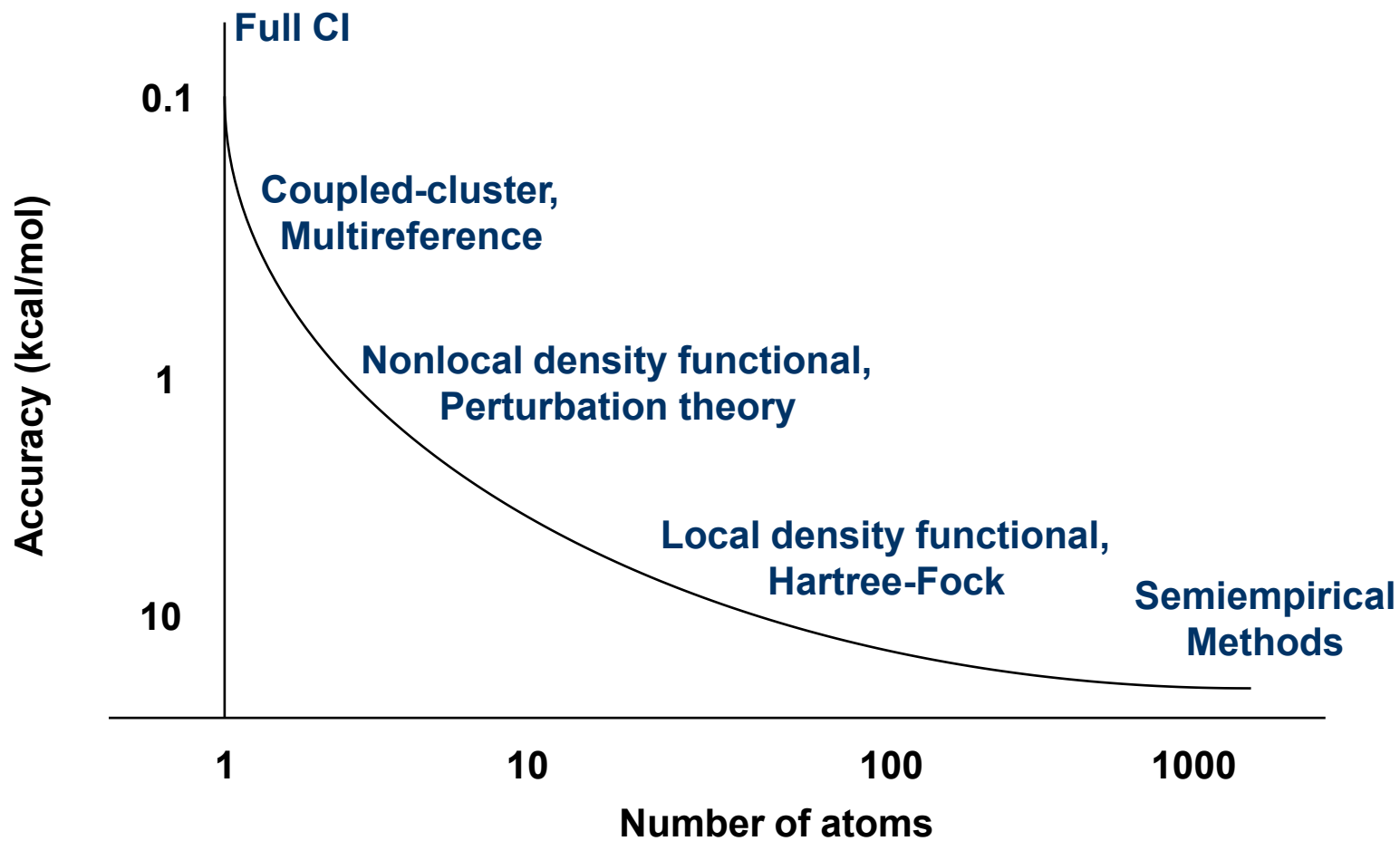
What should we learn?

- the theory behind “computational chemistry”
- to use program packages designed for molecular electronic structure theory
- to use some molecular visualization packages
- to do calculations and to interpret the results
- to make correlations between the experimental and theoretical data (QSAR)

Computational Chemistry



Size vs Accuracy



$$\hat{H} \Psi = E \Psi$$

Schrödinger equation

- Born-Oppenheimer approximation

$$\hat{H}_{el} \Psi_{el} = E \Psi_{el}$$

$$\phi_i = \sum_{\kappa}^M c_{\kappa i} \chi_{\kappa}$$

- Particle-independent approximation
- LCAO-MO approximation
- Slater-determinant

Hartree-Fock (HF-SCF) method

Ab initio

Semi-empirical

DFT

➤ Post HF methods

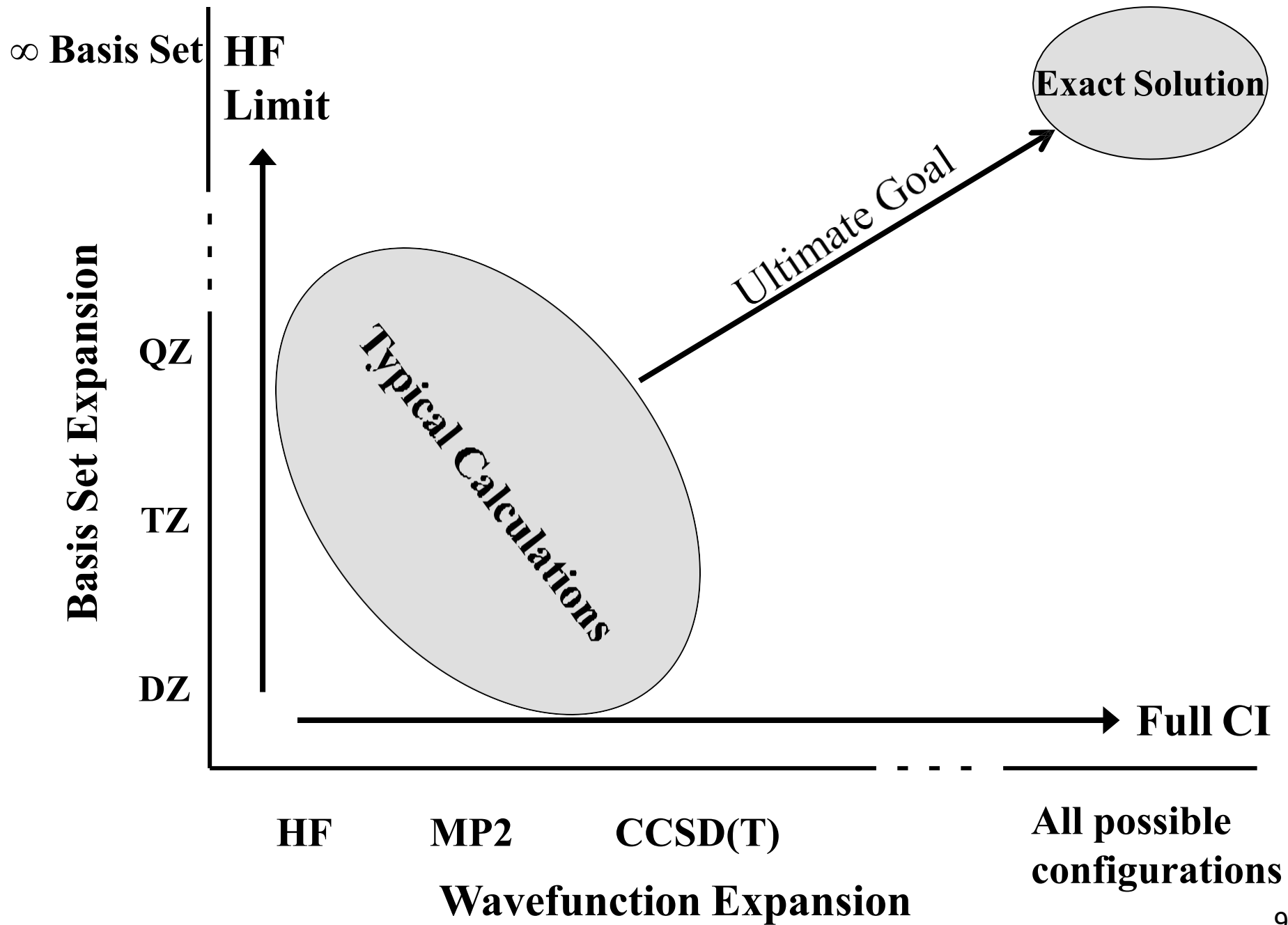
- **Perturbation Theory (MP2, MP3, MP4)**
- **Configuration Interaction (CI, MCSCF, CASSCF, MRCI, CCSD,...)**

- Neglect many of 2- $\bar{e}$ integrals
- Replace others by empirical parameters.
- **Models:**
CNDO, INDO, MINDO, NDDO, MNDO, AM1, PM3,...

Functional: “A function of a function”

Functional in DFT:
Energy $\equiv E[\rho]$

Density Functionals:
LDA, GGA, B3LYP, ...



Properties that can be calculated

- ❑ Equilibrium and transition-state structures,
- ❑ Dipole and higher multipole moments, and polarizabilities,
- ❑ Electronic charge distributions and spin density,
- ❑ Vibrational frequencies, IR and Raman Spectra,
- ❑ NMR spectra,
- ❑ Electronic excitations and UV-Vis spectra,
- ❑ MO energies and shapes (UPS and XPS spectra),
- ❑ Reaction mechanisms and kinetics,
- ❑ Thermochemical data,
- ❑ Reactivity,
- ❑ Conformational analysis
- ❑ Tautomerism,
- ❑ Structure-activity relationship, etc.

Getting into the theory...

- Three parts to solving the Schrödinger equation for molecules:
 - The Born-Oppenheimer Approximation
Leads to the idea of a potential energy surface
 - The Method
The expansion of the many-electron wave function in terms of Slater determinants.
 - The basis set
Representation of Slater determinants by molecular orbitals, which are linear combinations of atomic-like-orbital functions.

The Born-Oppenheimer Approximation

- The wave-function of the many-electron molecule is a function of electron and nuclear coordinates: $\psi(R,r)$ (R = nuclear coordinates, r = electron coordinates).
- The motions of the electrons and nuclei are coupled.
- However, the nuclei are much heavier than the electrons
 - $m_p \approx 2000 m_e$
- And consequently nuclei move ***much*** more slowly than do the electrons ($E=1/2mv^2$). To the electrons the nuclei appear fixed.
- Born-Oppenheimer Approximation: to a high degree of accuracy we can separate electron and nuclear motion:
$$\psi(R,r) = \psi_{el}(r;R) \psi_N(R)$$

Potential Energy Surfaces

- Structure-energy relationship is fundamental to computational chemistry
- the potential energy surface (PES) describes this relationship
- Now we can solve the electronic part of the Schrödinger equation separately.

$$\hat{H}_{el}\psi_{el}(r;R) = E_{el}\psi_{el}(r;R)$$

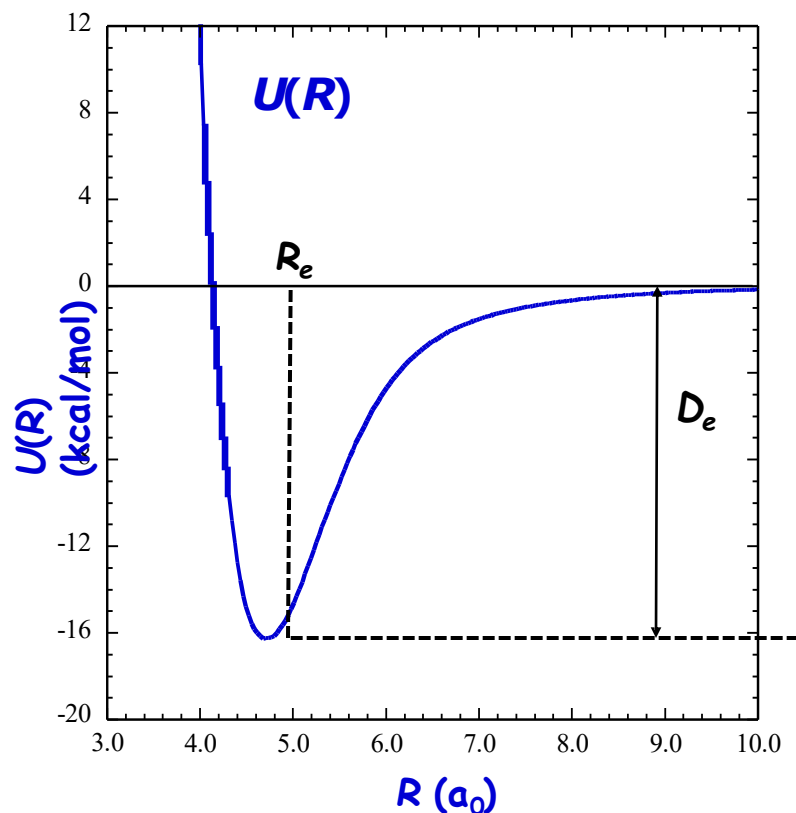
$$\hat{H}_{el} = -\frac{\hbar^2}{2m_e} \sum_i \nabla_i^2 - \sum_{\alpha} \sum_i \frac{Z_{\alpha} e'^2}{r_{i\alpha}} + \sum_j \sum_{i>j} \frac{e'^2}{r_{ij}}$$

- B-O approximation leads to the idea of a potential energy surface (PES).

$$U(R) = E_{el} + V_{NN}$$

$$V_{NN} = \sum_{\alpha} \sum_{\alpha>\beta} \frac{Z_{\alpha} Z_{\beta} e'^2}{r_{\alpha\beta}}$$

Diatomic Potential Energy Surface (HgBr)



Atomic unit of length
1 bohr = 1 a_0 = 0.529177 Å

Potential Energy Surfaces

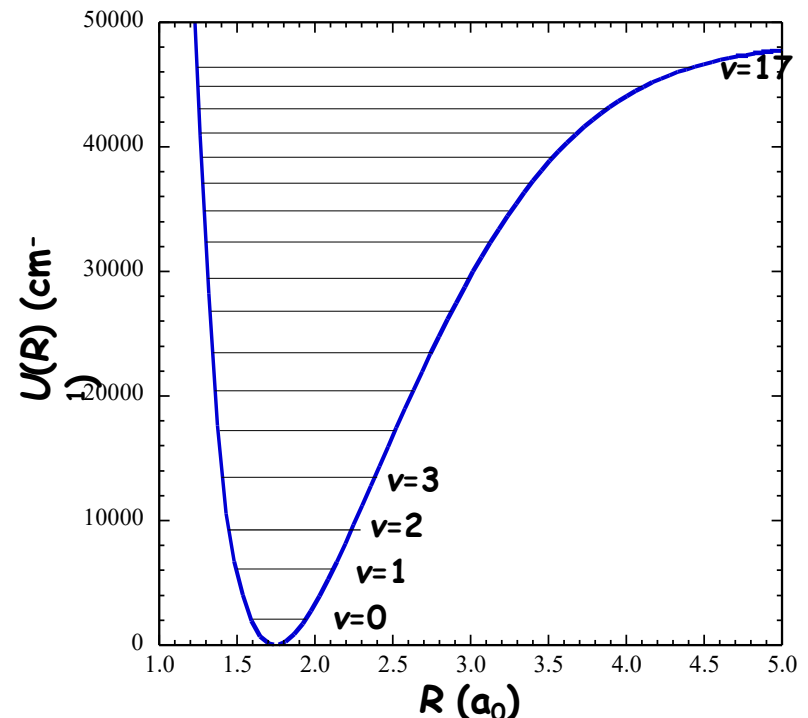
- Once we have the Potential Energy Surface (PES) we can solve the nuclear Schrödinger equation.

$$\hat{H}_N \psi_N(R) = E_N \psi_N(R)$$

$$\hat{H}_N = -\frac{\hbar^2}{2} \sum_{\alpha} \frac{1}{m_{\alpha}} \nabla_{\alpha}^2 + U(R)$$

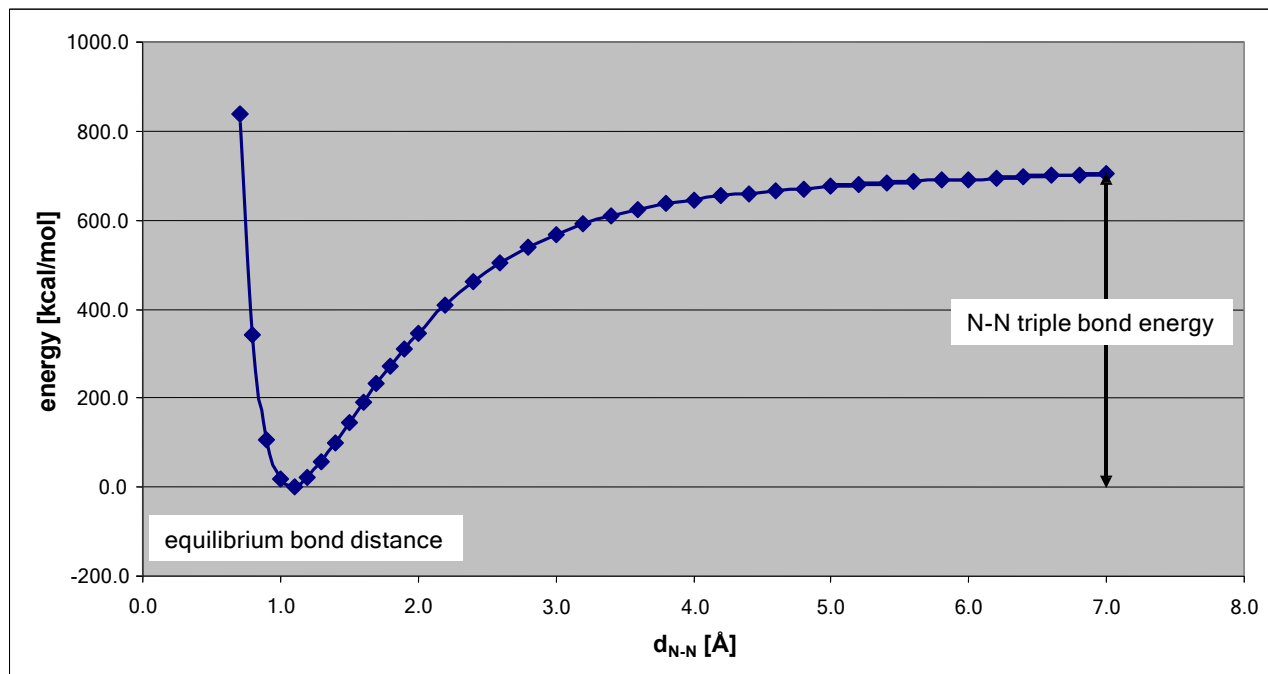
- Solution of the nuclear SE allow us to determine a large variety of molecular properties.
- An example are vibrational energy levels.

Vibrational Energy Levels of HF



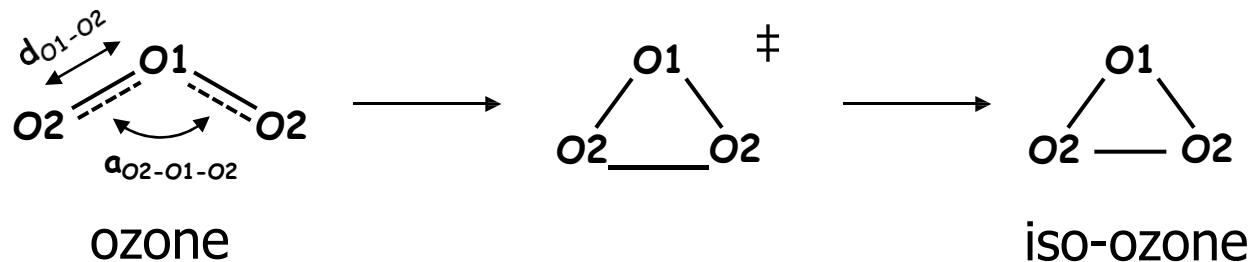
Potential Energy Surfaces

1-D Example: N₂



- calculate energy at several distinct values of d_{N-N}
 - called a potential energy surface scan
- interpolate between points to construct PES
- gives relevant structural and energetic information

2-D Example: Ozone Isomerization

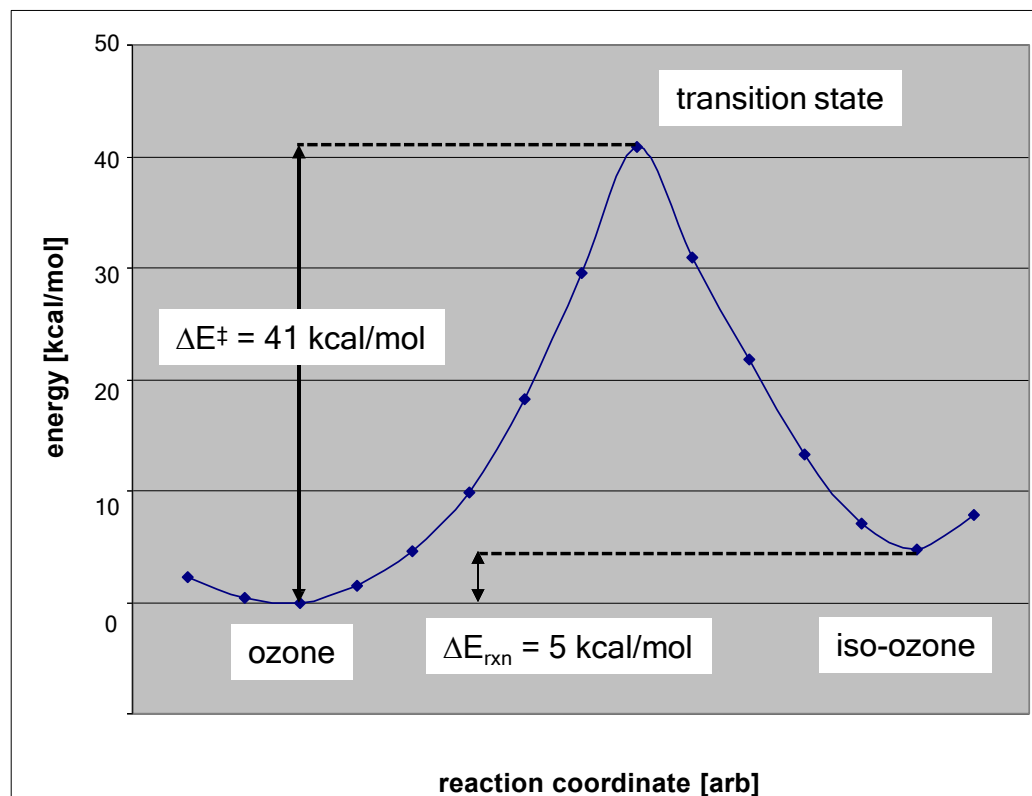


reaction coordinate:

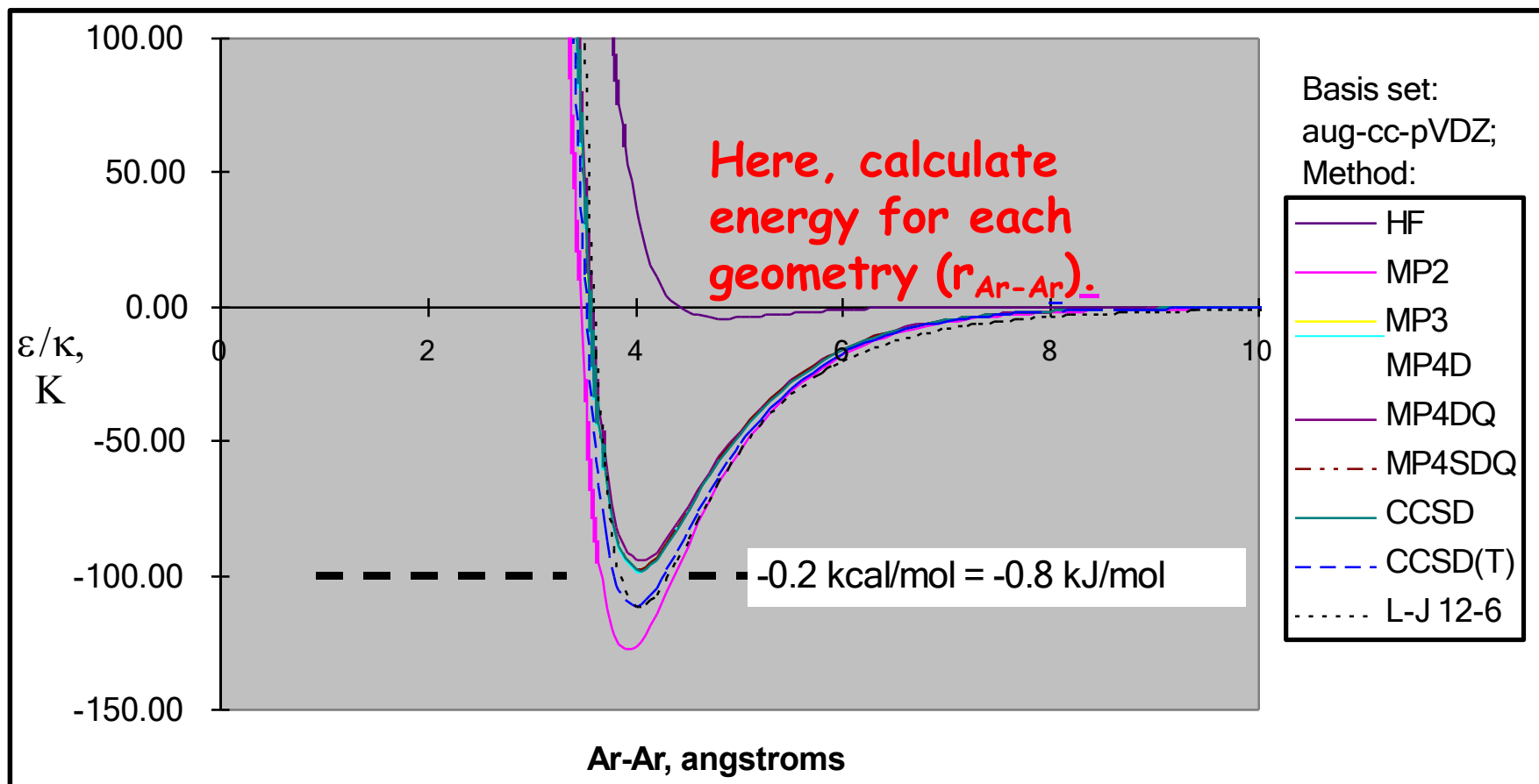
- lowest energy path on full PES connecting reactant, product, and transition state
- often plotted to aid in interpreting data

useful for identifying:

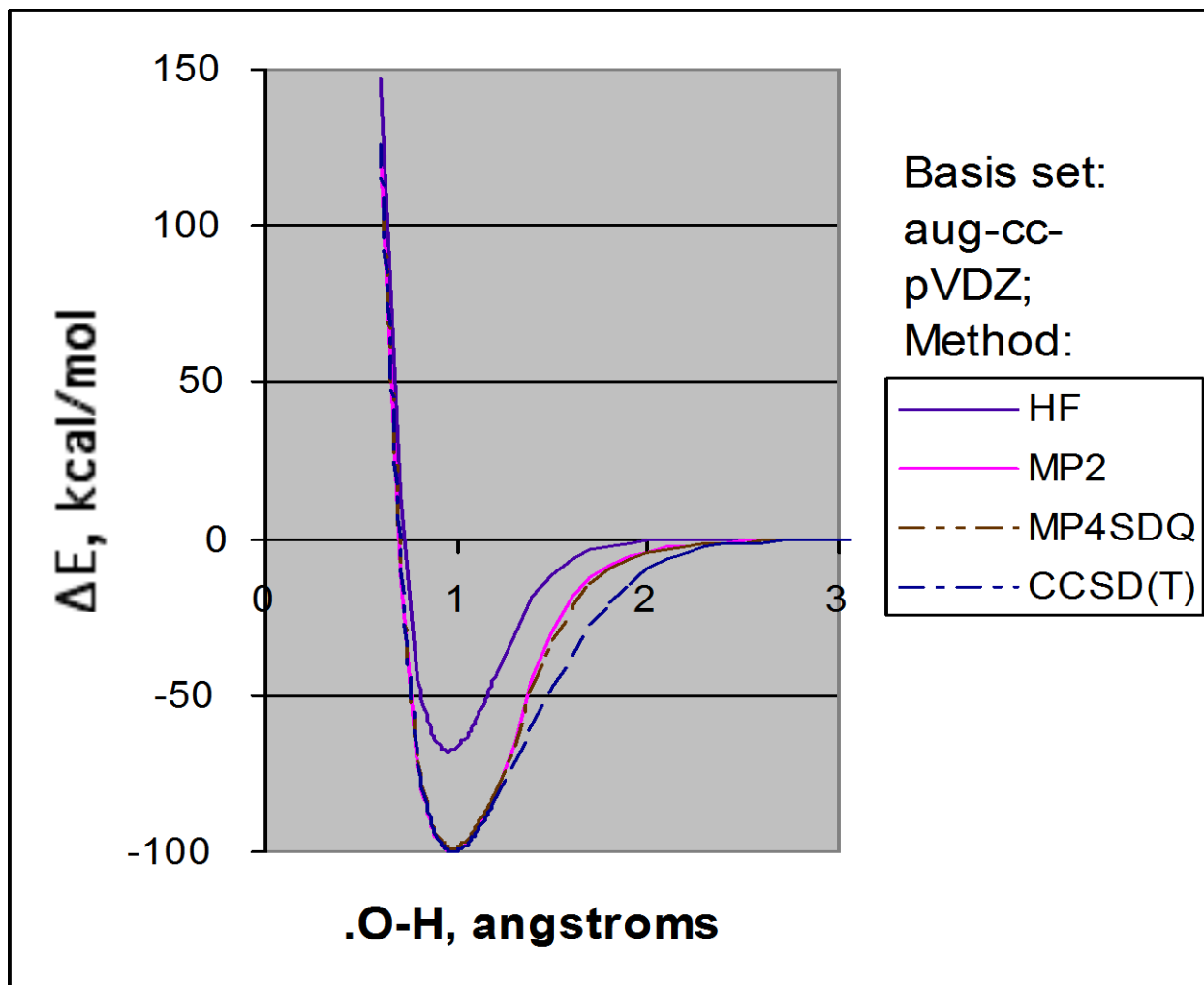
- reaction energies
- reaction barriers
- reactant, product and transition state geometries



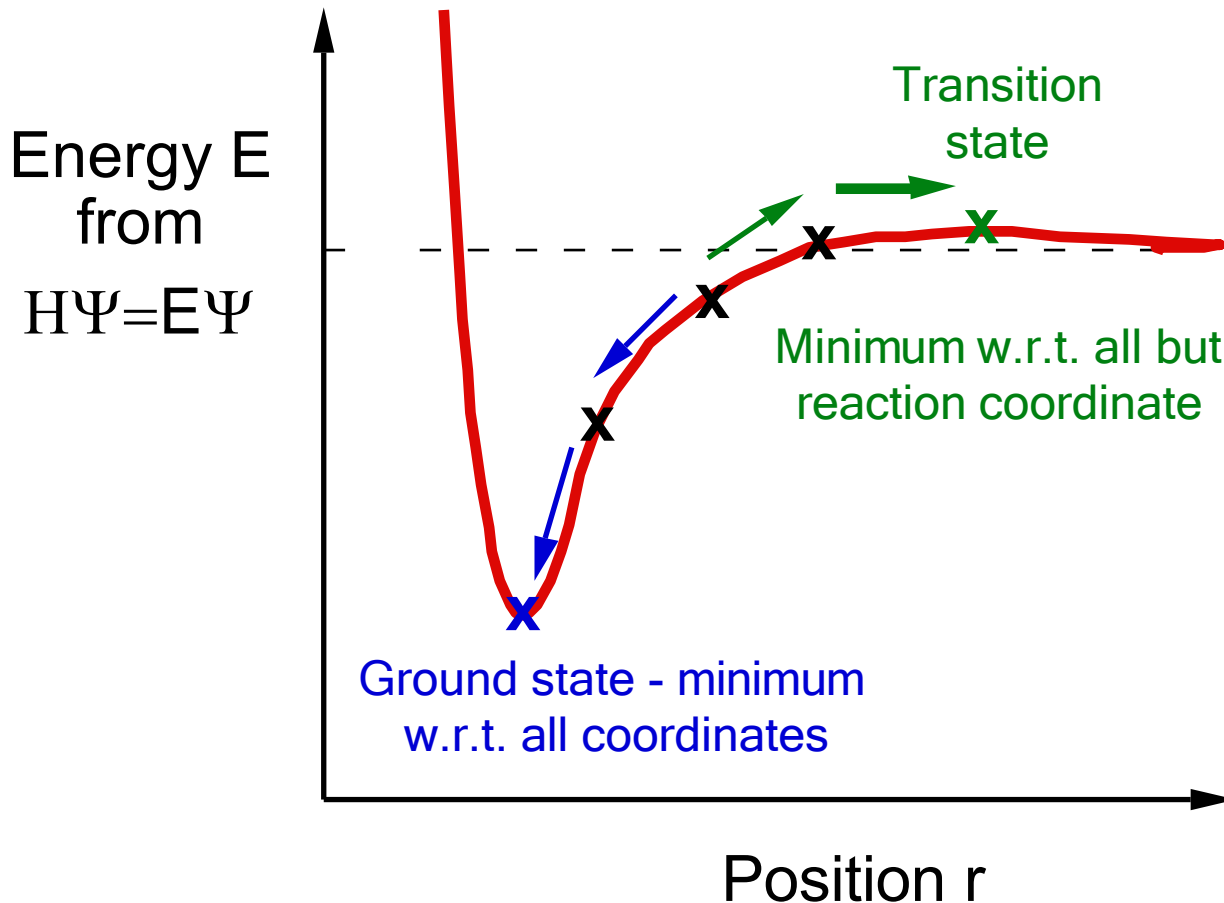
The van der Waals interaction energies well for an Ar dimer.



Chemical bond energies:



Geometry is found by optimizing computed energy with respect to coordinates.



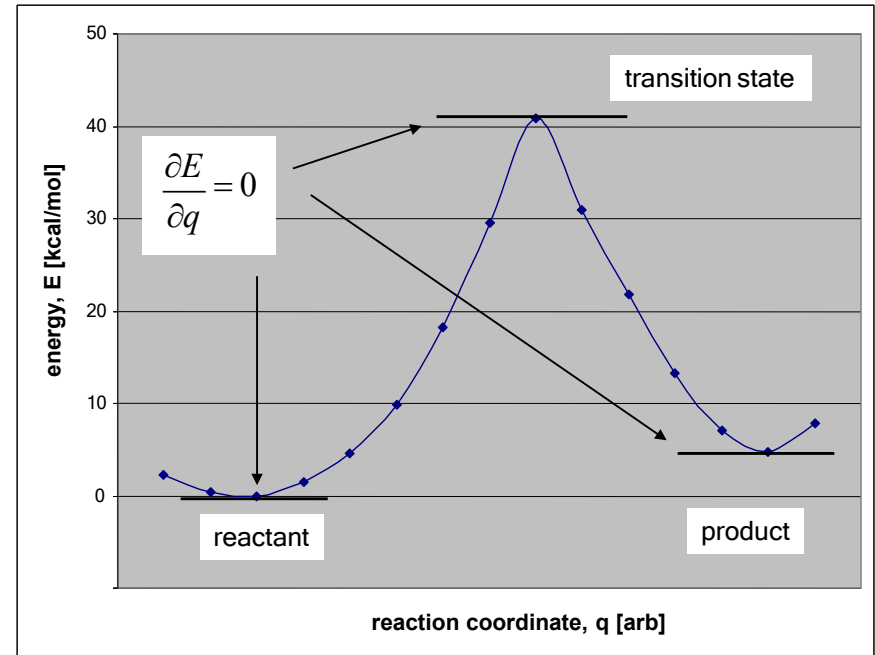
Stationary Points

reactants, products and transition states are stationary points

- mathematically, a stationary point is defined by:

$$\frac{\partial E}{\partial q_1} = \frac{\partial E}{\partial q_2} = \dots = \frac{\partial E}{\partial q_{3N-6}} = 0$$

along all directions, q_i



There are different kinds of stationary points

- reactants and products are local minima:
 - minimum in all directions
 - lowest energy minimum on the entire PES is called the global minimum
- transition states are saddle points:
 - maximum in only one direction, minima in all others

Stationary Points

Local minimum (reactant or product):

$$\frac{\partial E}{\partial q_i} = 0$$

- for all q_i
- stationary point

$$\frac{\partial^2 E}{\partial q_i^2} > 0$$

- for all q_i
- positive curvature in all directions

Saddle Point (transition state):

$$\frac{\partial E}{\partial q_i} = 0$$

- for all q_i
- stationary point

$$\frac{\partial^2 E}{\partial q_i^2} < 0$$

- for exactly one q_i
- negative curvature in one direction
- maximum along that direction

Second derivatives measure the rate of change in slopes (i.e. the curvature of the function f) near a critical point.

$$\frac{\partial^2 E}{\partial q_i^2} > 0$$

- for all other q_i
- positive curvature in all other directions
- minimum along all other directions

Geometry Optimization

- process for locating a minimum or transition state on the PES
- most common type of calculation performed in computational chemistry

Why?

- need geometries/energies of minima and transition states to:
 1. determine reaction mechanisms and associated energetics
 2. determine preferred molecular geometries for calculations of other properties

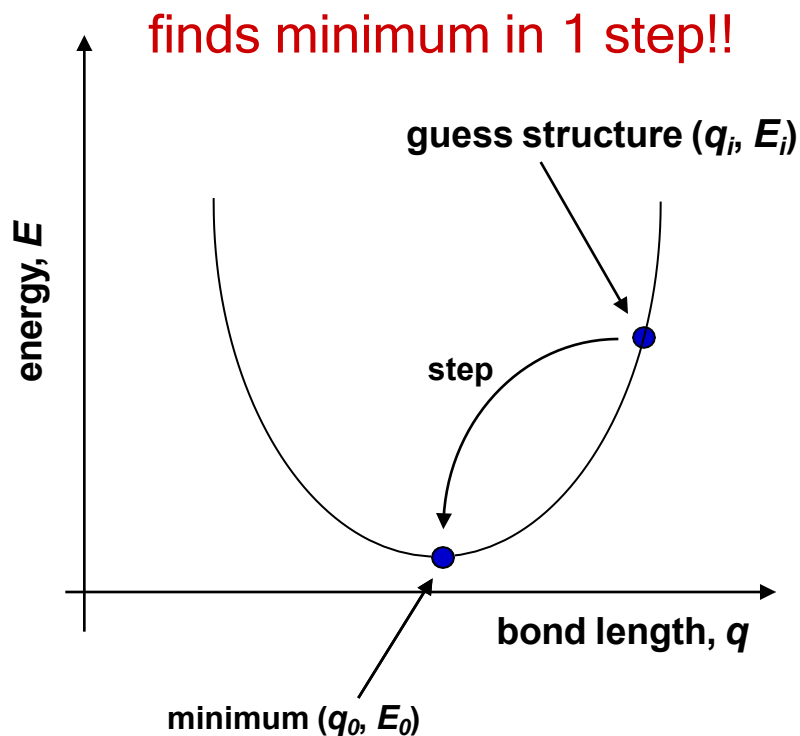
Basic idea:

- Iterative process
- start with a „guess“ structure and use information of the PES to change the coordinates such that the final structure is a stationary point
- lets us find minima/transition states without any exact information regarding those points on the PES

Newton-Raphson Method

- basis for geometry optimization procedures used in most computational chemistry codes

consider a 1-D harmonic potential



$$E_i - E_0 = \frac{1}{2} k (q_i - q_0)^2$$

$$\left(\frac{\partial E}{\partial q} \right)_i = k (q_i - q_0)$$

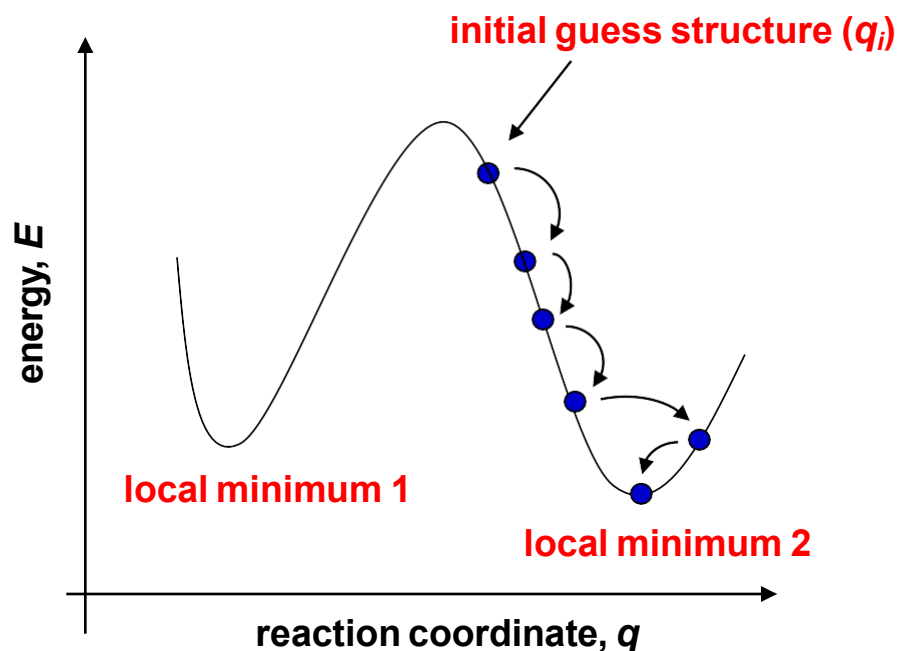
$$\left(\frac{\partial^2 E}{\partial q^2} \right)_i = k$$

$$\left(\frac{\partial E}{\partial q} \right)_i = \left(\frac{\partial^2 E}{\partial q^2} \right)_i (q_i - q_0)$$

$$q_0 = q_i - \left(\frac{\partial E}{\partial q} \right)_i / \left(\frac{\partial^2 E}{\partial q^2} \right)_i$$

Newton-Raphson Method

- in general, PES is not harmonic $\rightarrow$ cannot find q_0 in 1 step
- instead, we iterate until forces on nuclei (derivatives, gradients) are small
- with a reasonable guess structure, most programs will find the nearest local minimum in 10 - 20 iterations
- NOTE: doesn't just follow reaction coordinate, but optimizes over whole PES



$$q_{i+1} = q_i - \left(\frac{\partial E}{\partial q} \right)_i / \left(\frac{\partial^2 E}{\partial q^2} \right)_i$$
$$\downarrow$$
$$q_{i+2} = q_{i+1} - \left(\frac{\partial E}{\partial q} \right)_{i+1} / \left(\frac{\partial^2 E}{\partial q^2} \right)_{i+1}$$
$$\vdots$$
$$q_n = q_{n-1} - \left(\frac{\partial E}{\partial q} \right)_{n-1} / \left(\frac{\partial^2 E}{\partial q^2} \right)_{n-1}$$

Newton-Raphson Method

in general, optimization is done using 3N cartesian coordinates

$$q_{i+1} = q_i - \left(\frac{\partial E}{\partial q} \right)_i / \left(\frac{\partial^2 E}{\partial q^2} \right)_i \Rightarrow \mathbf{q}_{i+1} = \mathbf{q}_i - \mathbf{H}_i^{-1} \mathbf{g}_i$$

$$q_i \Rightarrow \mathbf{q}_i = \begin{bmatrix} q_{1i} \\ q_{2i} \\ \vdots \\ q_{3Ni} \end{bmatrix} \quad \left(\frac{\partial E}{\partial q} \right)_i \Rightarrow \mathbf{g}_i = \begin{bmatrix} (\partial E / \partial q_1)_i \\ (\partial E / \partial q_2)_i \\ \vdots \\ (\partial E / \partial q_{3N})_i \end{bmatrix}$$

$$\left(\frac{\partial^2 E}{\partial q^2} \right)_i \Rightarrow \mathbf{H}_i = \begin{bmatrix} (\partial^2 E / \partial q_1 q_1)_i & (\partial^2 E / \partial q_1 q_2)_i & \cdots & (\partial^2 E / \partial q_1 q_{3N})_i \\ (\partial^2 E / \partial q_2 q_1)_i & (\partial^2 E / \partial q_2 q_2)_i & \cdots & (\partial^2 E / \partial q_2 q_{3N})_i \\ \vdots & \vdots & \vdots & \vdots \\ (\partial^2 E / \partial q_{3N} q_1)_i & (\partial^2 E / \partial q_{3N} q_2)_i & \cdots & (\partial^2 E / \partial q_{3N} q_{3N})_i \end{bmatrix}$$

Newton-Raphson Method

to do an optimization, we need:

- q_i
 - column vector of 3N cartesian coordinates defining the molecular geometry
 - initial guess structure should look like anticipated minimum or transition state
 - can build with molecular editors
- g_i
 - column vector of 3N derivatives of the energy wrt cartesian coordinates
 - represent the forces on the nuclei: $F_a = -\partial E / \partial q_a$
 - calculated by the program $\rightarrow$ ~20% of effort in geometry optimization
- H_i
 - “Hessian” matrix containing 3Nx3N 2nd derivatives of the energy w.r.t. cartesian coordinates
 - 2nd derivatives are expensive to calculate analytically
 - generally estimated during optimization:

$$\left(\frac{\partial^2 E}{\partial q_a \partial q_b} \right)_i \approx \frac{\Delta(\partial E / \partial q_b)_i}{\Delta q_a}$$

Finding Minima

1. construct a “guess” structure → optimization will go to nearest minimum
2. build up initial Hessian, H_i (approximate from low-level calculation)
3. calculate derivatives, g_i
4. compare derivatives against convergence criteria
5. update geometry: $\mathbf{q}_{i+1} = \mathbf{q}_i - \mathbf{H}_i^{-1} \mathbf{g}_i$
6. update Hessian:
$$\left(\frac{\partial^2 E}{\partial q_a \partial q_b} \right)_i \approx \frac{\Delta(\partial E / \partial q_b)_i}{\Delta q_a}$$
7. repeat 3 through 6 until convergence criteria are met

Convergence Criteria

- impossible to find exact location of minimum
- instead geometry optimization stops when all derivatives are small
- in most codes:
 - ✓ maximum force $< 4.5 \times 10^{-4}$ Hartree/bohr
 - ✓ RMS force $< 3.0 \times 10^{-4}$ Hartree/bohr
 - ✓ other geometry-based criteria

Finding Transition States

1. construct a “guess” structure → need good guess for transition states
2. build up initial Hessian, H_i (calculate analytically)
3. calculate derivatives, g_i
4. compare derivatives against convergence criteria
5. update geometry: $\mathbf{q}_{i+1} = \mathbf{q}_i - \mathbf{H}_i^{-1} \mathbf{g}_i$ (modify H for TS)
6. update Hessian:

$$\left(\frac{\partial^2 E}{\partial q_a \partial q_b} \right)_i \approx \frac{\Delta(\partial E / \partial q_b)_i}{\Delta q_a}$$

7. repeat 3 through 6 until convergence criteria are met

Initial guess

- finding a saddle point is difficult → need good guess structure, accurate Hessian
- guess structure by:
 - preliminary scan along reaction coordinate
 - interpolate between reactant and product structures
 - chemical intuition

Characterizing Stationary Points

- convergence criteria are based only on derivatives
- definitions of minima and transition states involve 2nd derivatives:

minimum: $\frac{\partial^2 E}{\partial q_i^2} > 0$ for all q_i

transition state: $\frac{\partial^2 E}{\partial q_i^2} < 0$ for exactly one q_i

$$\frac{\partial^2 E}{\partial q_i^2} > 0 \quad \text{for all other } q_i$$

need 2nd derivatives to characterize minima/transition states

→ frequency calculation and normal mode analysis

Characterizing Stationary Points

minimum: $\frac{\partial^2 E}{\partial q_i^2} > 0$ for all q_i

- all positive elements in $H \rightarrow$ all diagonal elements of K are positive

$$\tilde{\nu} = \frac{1}{2\pi c} \left(\frac{k}{\mu} \right)^{1/2} \rightarrow \text{if } k \text{ is positive, the frequency is positive}$$

minima are characterized as having all positive frequencies

transition state: $\frac{\partial^2 E}{\partial q_i^2} < 0$ for exactly one q_i

- one negative element in $H \rightarrow$ one diagonal element of K is negative

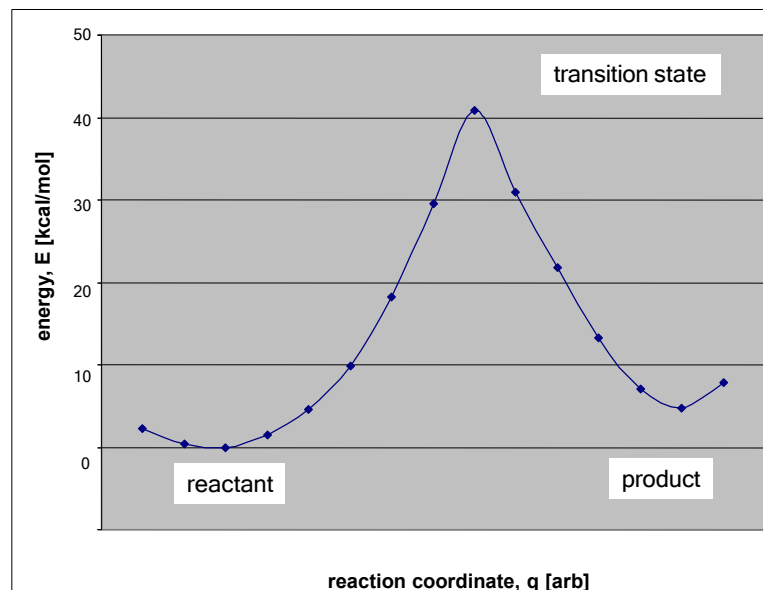
$$\tilde{\nu} = \frac{1}{2\pi c} \left(\frac{k}{\mu} \right)^{1/2} \rightarrow \text{if } k \text{ is negative, the frequency is imaginary}$$

transition states are characterized as having exactly one imaginary frequency

Characterizing Stationary Points

How do you know you have the correct transition state?

1. Look at the structure. Does it look right?
 - pretty weak confirmation method, but helpful sometimes
2. The energy of the transition state must be higher than the reactant and product.



Characterizing Stationary Points

How do you know you have the correct transition state?

1. Look at the structure. Does it look right?
 - pretty weak confirmation method, but helpful sometimes
2. The energy of the transition state must be higher than the reactant and product.
3. Animate the normal mode with the imaginary frequency
 - the normal mode with the imaginary frequency corresponds to the motion of the molecule over the energy barrier. Does this motion connect the reactants to the products?
4. Perform an intrinsic reaction coordinate (IRC) calculation
 - IRC calculations start at the transition state and follow the lowest energy pathway in each direction along the imaginary normal mode. Do these pathways lead to the reactants and products?
 - IRC calculations are expensive and rarely necessary. But, they are a surefire way to characterize the transition state.

Relative Energies

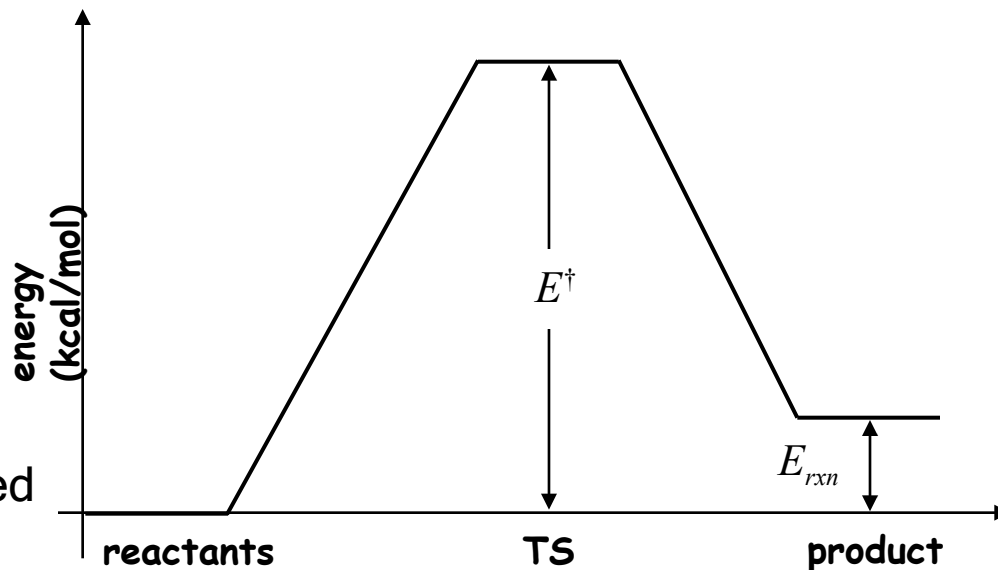
once you identify the reactant, product, and transition state

- energies can be calculated relative to reactant

$$E_{rxn} = E_{prod} - E_{react}$$

$$E^{\ddagger} = E_{TS} - E_{react}$$

- sets reactant energy to 0
- all energies must be calculated at the same level of theory!!!



For reactions with multiple reactants or products:

e.g. $A + B \rightarrow TS^{\ddagger} \rightarrow C + D$

$$E_{rxn} = \sum E_{prod} - \sum E_{react}$$

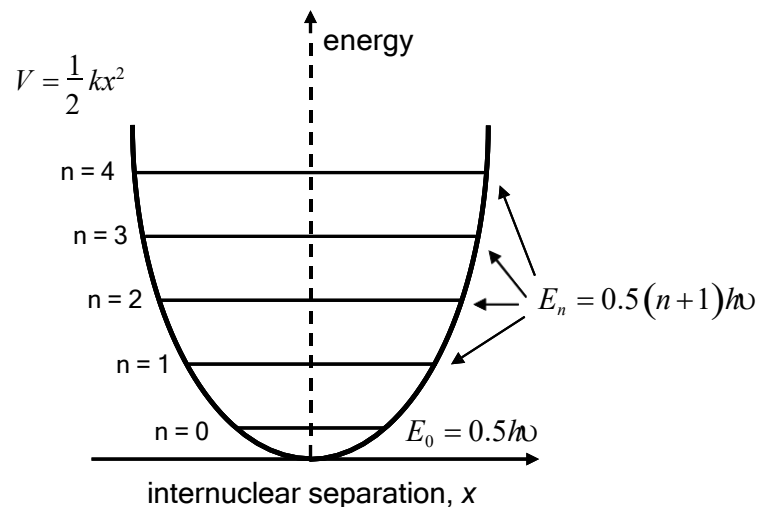
$$E^{\ddagger} = E_{TS} - \sum E_{react}$$

- energies are calculated separately for each molecule
- e.g. do a calculation on A and a calculation on B, not a calculation on A and B together

Zero-Point Vibrational Energies

Quantum mechanics:

- molecules vibrate even at 0K
 - Heisenberg uncertainty principle says we can't know simultaneously the position and velocity of a particle
 - if atoms stopped vibrating they would have no velocities and well-defined positions



Zero-Point Vibrational Energy:

- vibrational energy with $n = 0$
- added to potential energy of the molecule

$$E_{tot}(\{R_I\}) = E(\{R_I\}) + 0.5h \sum_i \nu_i$$

potential energy from
an energy calculation

zero-point vibrational
energy from normal
mode calculation

n = vibrational state

h = Planck's constant
= $6.626 \times 10^{-34} \text{ Js}^{-1}$

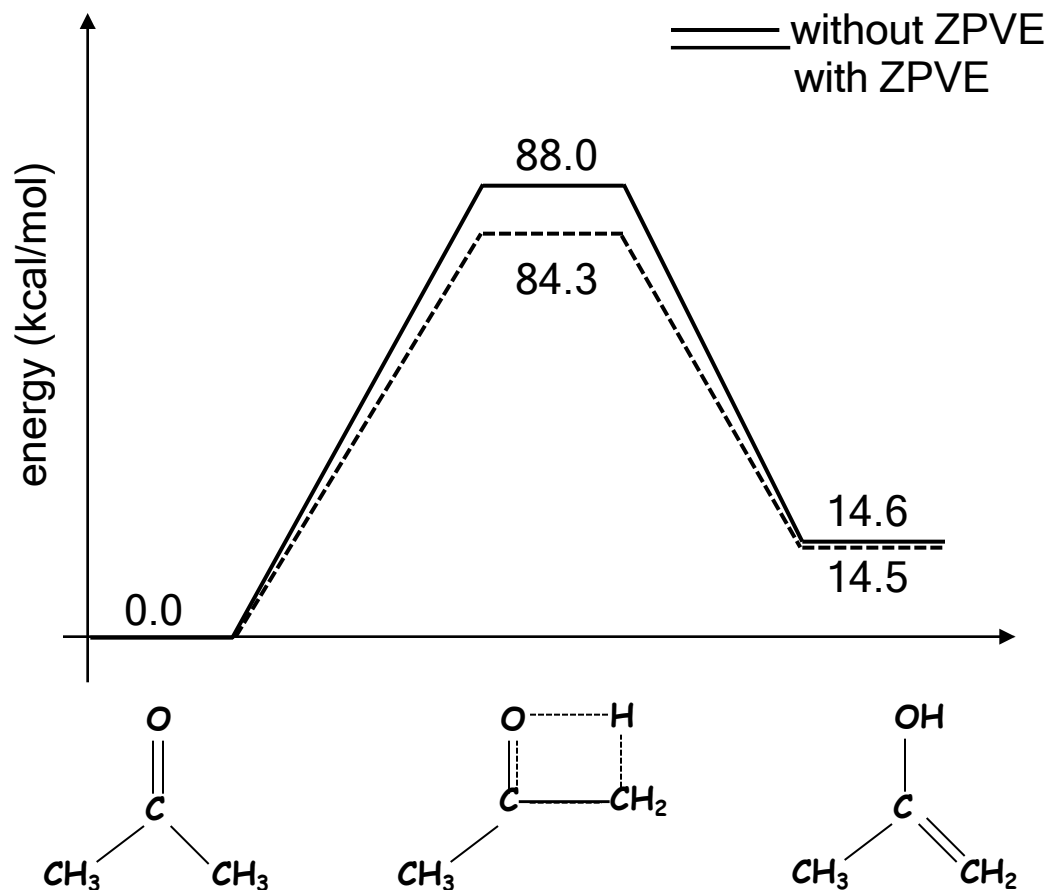
ν_i = frequencies from normal
mode calculation

Zero-Point Vibrational Energies

zero-point vibrational energy has small effect on relative energies

$$E_{tot}(\{R_I\}) = E(\{R_I\}) + 0.5h \sum_i \nu_i$$

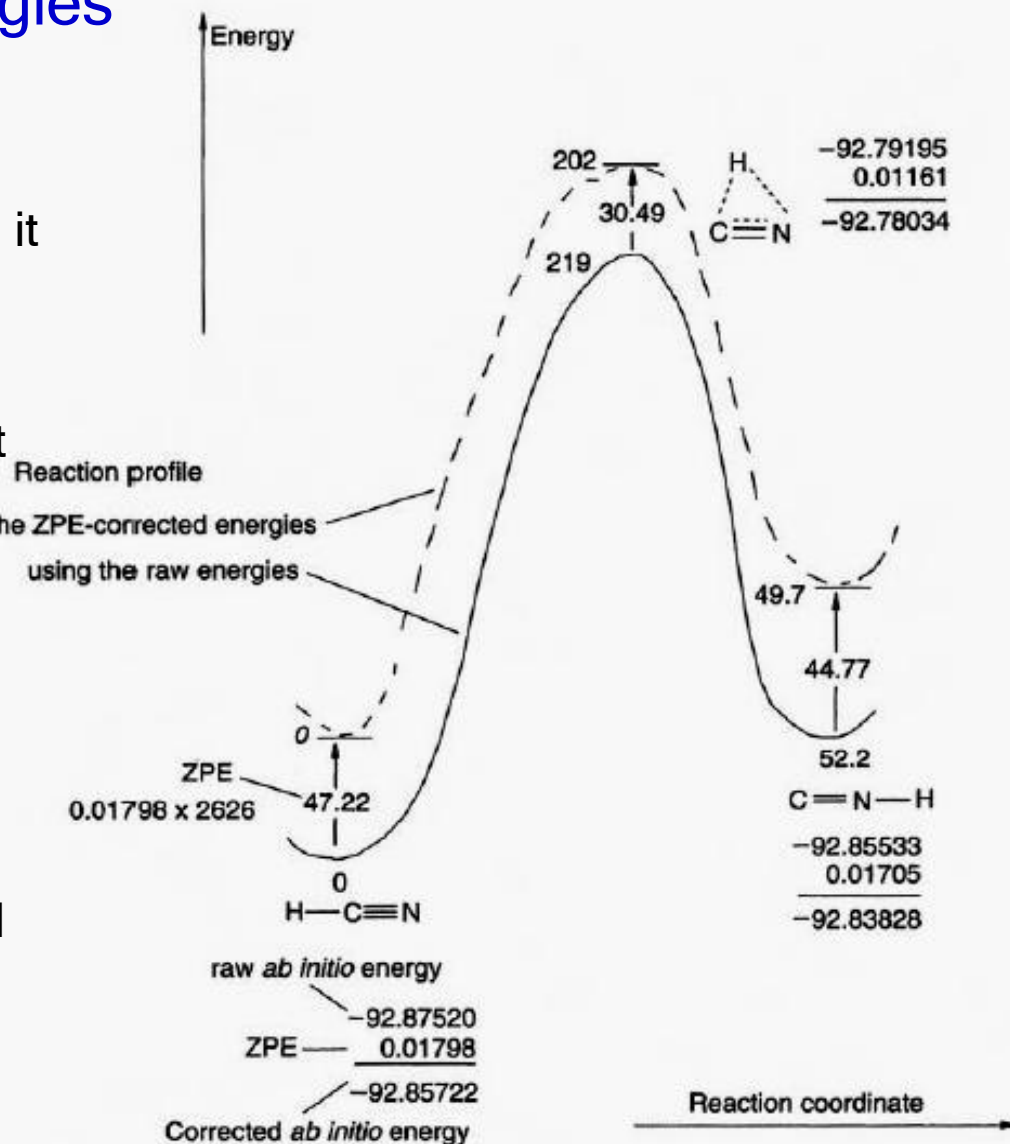
- nearly cancels out for reaction energies
- effect is often more pronounced for barriers
- transition states often have many weak vibrations with low frequencies
- ZPVE correction is smaller than for reactants and products



- *often included in reaction energies and barriers if frequencies are available*

Zero-Point Vibrational Energies

- The ZPE is the energy a molecule has even at absolute zero because it still vibrates.
- The ZPE of a species is usually not small compared to activation energies or reaction energies,
- For accurate work the ZPE should be added to the total energies of species and the ZPE-corrected energies should then be compared



Thermal Corrections

Energy calculation methods provide the potential energy of the system at 0K

Meanwhile, experiments:

- are done at finite temperatures and pressure
- are governed by enthalpies and free energies

how do we incorporate thermal and pressure effects into our calculated energies?

- use statistical mechanics to explore the properties of the system in an macroscopic ensemble
- statistical mechanics will let us determine how thermal energy is distributed in the molecule
- provides access to enthalpies, entropies, and free energies

Relative Enthalpies/Free Energies

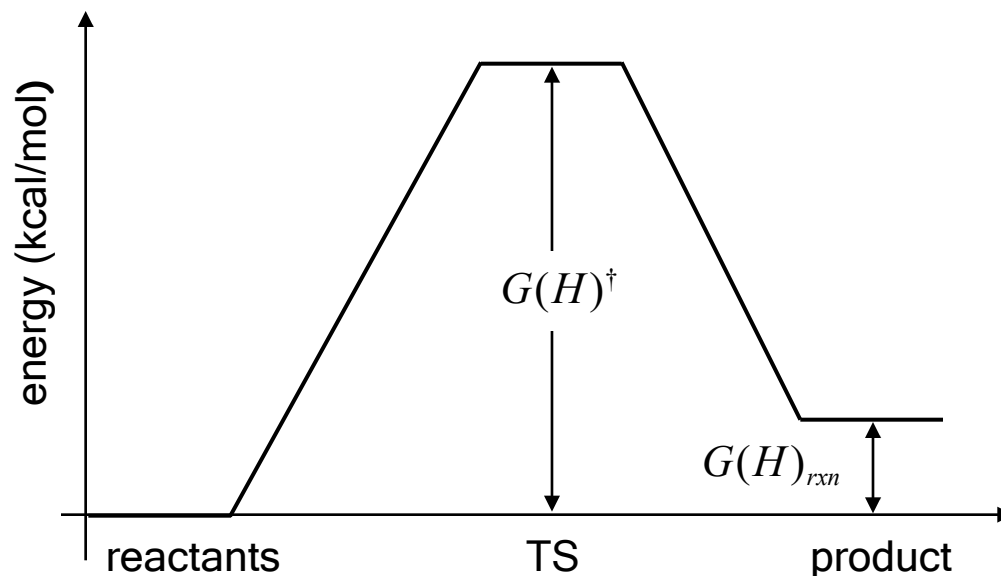
once you identify the reactant, product, and transition state

- enthalpies/free energies can be calculated relative to reactant

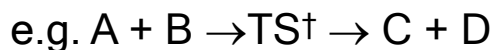
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$$G(H)^{\dagger} = G(H)_{TS} - G(H)_{react}$$

- sets reactant energy to 0
- all energies must be calculated at the same level of theory!!!



For reactions with multiple reactants or products:



$$G(H)_{rxn} = \sum G(H)_{prod} - \sum G(H)_{react}$$

$$G(H)^{\dagger} = G(H)_{TS} - \sum G(H)_{react}$$

- energies are calculated separately for each molecule
- e.g. do a calculation on A and a calculation on B, not a calculation on A and B together

Summary of PES-related Concepts

1. Potential energy surface (PES)

- describes relationship between molecular structure and energy
- PES evaluated with energy calculation methods (future lectures)

2. Stationary Points

- points on the PES where $\frac{\partial E}{\partial q_1} = \frac{\partial E}{\partial q_2} = \dots = \frac{\partial E}{\partial q_{3N-6}} = 0$
- reactants and products are local minima (all positive 2nd derivatives)
- transition states are saddle-points (exactly one negative 2nd derivative)

3. Geometry Optimization

- procedure for identifying stationary points on the PES

4. Normal Mode Analysis/Frequency Calculation

- relate vibrational frequencies to 2nd of energy wrt coordinates
- can be used to characterize stationary points and calculate ZPVEs
- only valid at stationary points
- provide access to enthalpies, entropies and free energies in ideal gas approx.

Three key features of theory are required for *quantum chemical* calculations.

- Understand how initial specification of nuclear positions is used to calculate energy:
 - Solving the Schrödinger equation
- Understand “basis sets” and how to choose them:
 - Functions that represent the atomic orbitals
 - e.g., 3-21G, 6-311++G(3df,2pd), cc-pVTZ
- Understand levels of theory and how to choose them:
 - Wavefunction methods: Hartree-Fock, MP4, CI, CAS
 - Density functional methods: LYP, B3LYP, etc.
 - Compound methods: CBS, G3
 - Semiempirical methods: AM1, PM3
- Then, calculate properties.